Piecewise-Defined & Implicit Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating piecewise functions**

A shipping company charges a flat 5 dollars for packages up to 2 kg, plus 1.50 dollars per kg for anything over 2 kg. Write a piecewise function $C(w)$ for the cost, and find $C(5)$.

$C(w) = 5$ for $0 < w \leq 2$, and $C(w) = 5 + 1.5(w - 2)$ for $w > 2$. $C(5) = 5 + 1.5(5 - 2) = 5 + 4.5 = 9.5$, so 9.5 dollars.

2 **[Foundational]** • **Target: Checking continuity of piecewise functions**

Find the value of k that makes $f(x) = 3x + k$ for $x < 2$, and $f(x) = x^2 - 1$ for $x \geq 2$, continuous at $x = 2$.

For continuity, the two pieces must agree at $x = 2$: $3(2) + k = 2^2 - 1$, so $6 + k = 3$, giving $k = -3$.

3 **[Foundational]** • **Target: Checking continuity of piecewise functions**

Determine whether $g(x) = x + 1$ for $x \leq 1$, and $g(x) = 3x - 2$ for $x > 1$, is continuous at $x = 1$, showing your reasoning.

Left piece at $x = 1$: $1 + 1 = 2$. Right piece as $x \to 1^+$: $3(1) - 2 = 1$. Since $2 \neq 1$, the two pieces do not match at $x = 1$, so g is not continuous there — there is a jump discontinuity.

4 **[Foundational]** • **Target: Visualizing a piecewise graph**

The graph shows $f(x) = -x - 1$ for $x < 0$, and $f(x) = x^2$ for $x \geq 0$.

a Evaluate the left piece and the right piece of f at $x = 0$, and use this to determine whether f is continuous there. Left piece as $x \to 0^-$: $-0 - 1 = -1$. Right piece at $x = 0$: $0^2 = 0$. Since $-1 \neq 0$, f has a jump discontinuity at $x = 0$, visible as the gap between the two branches in the graph.

b State the range of f shown on the graph. The linear branch for $x < 0$ takes all values greater than -1 (approaching but not including -1), and the parabola branch for $x \geq 0$ takes all values ≥ 0. Combined range: $(-1, \infty)$.

(a) Left piece as $x \to 0^-$: $-0 - 1 = -1$. Right piece at $x = 0$: $0^2 = 0$. Since $-1 \neq 0$, f has a jump discontinuity at $x = 0$, visible as the gap between the two branches in the graph.; (b) The linear branch for $x < 0$ takes all values greater than -1 (approaching but not including -1), and the parabola branch for $x \geq 0$ takes all values ≥ 0. Combined range: $(-1, \infty)$.

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5 [Foundational] • Target: Implicit relations and testing for functions

Determine whether $x^2 + y^2 = 25$ defines y as a function of x. Justify using the vertical line test.

This is a circle of radius 5. A vertical line such as $x = 3$ intersects it at two points ($y = 4$ and $y = -4$), so it fails the vertical line test — y is not a function of x.

6 [Foundational] • Target: Implicit relations and testing for functions

Solve $x^2 + y^2 = 25$ for y explicitly, and state which branch corresponds to a function.

$y = \pm\sqrt{25 - x^2}$. Each branch individually is a function: $y = \sqrt{25 - x^2}$ (upper semicircle) or $y = -\sqrt{25 - x^2}$ (lower semicircle) — together they form the full circle, which is not a function.

7 [Foundational] • Target: Implicit differentiation-style reasoning (algebraic slope)

For the implicit relation $x^2 + 4y^2 = 16$ (an ellipse), find the two y-values when $x = 2$, and describe what this means about the relation as a function.

$4 + 4y^2 = 16$, so $4y^2 = 12$, $y^2 = 3$, giving $y = \pm\sqrt{3}$. Since $x = 2$ produces two distinct y-values, the relation is not a function of x over its full domain.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Modeling with piecewise functions

A tax schedule charges 10% on income up to 10{,}000 dollars and 20% on income above 10{,}000 dollars. Write a piecewise tax function $T(x)$ for income x, and find the tax owed on an income of 25{,}000 dollars.

$T(x) = 0.10x$ for $0 \le x \le 10000$, and $T(x) = 1000 + 0.20(x - 10000)$ for $x > 10000$.
$T(25000) = 1000 + 0.20(15000) = 1000 + 3000 = 4000$, so 4000 dollars in tax.

9 [Intermediate] • Target: Piecewise functions with three or more pieces

A parking garage charges 3 dollars for the first hour, 2 dollars for each additional hour up to 5 hours total, and a flat 15 dollars for anything over 5 hours. Write this as a three-piece function $P(h)$ for h hours parked, and find $P(4)$.

$P(h) = 3$ for $0 < h \le 1$; $P(h) = 3 + 2(h - 1)$ for $1 < h \le 5$; $P(h) = 15$ for $h > 5$.
$P(4) = 3 + 2(4 - 1) = 3 + 6 = 9$, so 9 dollars.

10 [Intermediate] • Target: Piecewise functions with three or more pieces

Evaluate the absolute value function $f(x) = |2x - 6|$ at $x = 1$ and $x = 5$ by first rewriting it as a two-piece function without absolute value bars.

Since $2x - 6 = 0$ at $x = 3$: $f(x) = -(2x - 6) = 6 - 2x$ for $x < 3$, and $f(x) = 2x - 6$ for $x \ge 3$.
$f(1) = 6 - 2(1) = 4$. $f(5) = 2(5) - 6 = 4$.

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11 **[Intermediate]** • **Target: Graphing implicit relations point-by-point**

For the implicit relation $y^2 = x + 4$, find all y-values when $x = 0$, $x = 5$, and $x = -4$, and state the smallest allowed x-value.

At $x = 0$: $y^2 = 4$, so $y = \pm 2$. At $x = 5$: $y^2 = 9$, so $y = \pm 3$. At $x = -4$: $y^2 = 0$, so $y = 0$ (a single point). Since $y^2 \geq 0$ requires $x + 4 \geq 0$, the smallest allowed x-value is $x = -4$.

12 **[Intermediate]** • **Target: Piecewise functions and average rate of change**

For $f(x) = x^2$ on $x < 2$ and $f(x) = 3x - 2$ on $x \geq 2$, find the average rate of change of f on $[0, 4]$.

$f(0) = 0^2 = 0$ (using the first piece). $f(4) = 3(4) - 2 = 10$ (using the second piece, since $4 \geq 2$). AROC $= \frac{10 - 0}{4 - 0} = \frac{10}{4} = 2.5$.

13 **[Intermediate]** • **Target: Piecewise functions and average rate of change**

Given $g(x) = 2x + 1$ for $x \leq 3$ and $g(x) = x^2 - 5$ for $x > 3$, find the average rate of change of g on $[1, 5]$.

$g(1) = 2(1) + 1 = 3$ (first piece, since $1 \leq 3$). $g(5) = 5^2 - 5 = 20$ (second piece, since $5 > 3$). AROC $= \frac{20 - 3}{5 - 1} = \frac{17}{4} = 4.25$.

14 **[Intermediate]** • **Target: Piecewise functions and average rate of change**

A cell phone plan charges 20 dollars flat for up to 2 GB of data, plus 8 dollars per GB for each additional GB, up to a maximum monthly charge of 68 dollars (unlimited data cap). Write a three-piece function $M(x)$ for the monthly cost based on x GB used, and find $M(8)$.

$M(x) = 20$ for $0 \leq x \leq 2$; $M(x) = 20 + 8(x - 2)$ for $2 < x \leq 8$ (since $20 + 8(6) = 68$ is reached exactly at $x = 8$); $M(x) = 68$ for $x > 8$. $M(8) = 20 + 8(8 - 2) = 20 + 48 = 68$, so 68 dollars.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Modeling with piecewise functions** • **Cross-Lesson: 2.02 Logarithmic Functions & Properties**

If $\sin\theta = -\frac{4}{5}$ and θ lies in Quadrant III, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

Using the Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$: $\left(-\frac{4}{5}\right)^2 + \cos^2\theta = 1 \Rightarrow \frac{16}{25} + \cos^2\theta = 1 \Rightarrow \cos^2\theta = \frac{9}{25}$. Since θ is in Quadrant III, cosine is negative: $\cos\theta = -\frac{3}{5}$. Tangent is positive in Quadrant III: $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{-4/5}{-3/5} = \frac{4}{3}$. Double-angle formula for sine.

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16 **[Challenging]** • **Target: Piecewise functions with three or more pieces** • **Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle**

Let $f(x) = \frac{x^2 - 9}{x^2 - 5x + 6}$. Identify the locations of all vertical asymptotes and removable discontinuities (holes) of $f(x)$.

Factor numerator and denominator completely: $f(x) = \frac{(x-3)(x+3)}{(x-2)(x-3)}$ Notice that $(x - 3)$ is a common factor in both numerator and denominator: Removable Discontinuity (Hole): Occurs at $x = 3$. To find the y -coordinate, evaluate the simplified fraction: $f_{\text{reduced}}(3) = \frac{3+3}{3-2} = \frac{6}{1} = 6$. The hole is at **(3, 6)**.Vertical Asymptote: The remaining uncancelled denominator factor is $x - 2 = 0$, creating a vertical asymptote at line **x = 2**.

17 **[Challenging]** • **Target: Graphing implicit relations point-by-point** • **Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs**

Find all real and complex roots of the polynomial $P(x) = 2x^3 + 3x^2 - 8x + 3$.

By the Rational Root Theorem, possible rational roots are $\pm\{1, 3\}/\{1, 2\} = \pm\{1, 3, 1/2, 3/2\}$. Test $x = 1$: $P(1) = 2(1)^3 + 3(1)^2 - 8(1) + 3 = 2 + 3 - 8 + 3 = 0$. ✓ So $(x - 1)$ is a factor. Divide $P(x)$ by $(x - 1)$ using synthetic division: $2x^3 + 3x^2 - 8x + 3 = (x - 1)(2x^2 + 5x - 3)$. Factor the remaining quadratic: $2x^2 + 5x - 3 = (2x - 1)(x + 3)$. Setting each factor to zero yields the complete set of roots: $x = 1, \quad x = \frac{1}{2}, \quad x = -3$.

18 **[Challenging]** • **Target: Piecewise functions and average rate of change** • **Cross-Lesson: 4.01 Inverse & Composite Functions**

Evaluate the limit $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$ algebraically, and interpret what happens to the graph of $f(x) = \frac{x^2 - 9}{x - 3}$ at $x = 3$.

Step 1 (Factor and Cancel): Direct substitution gives $\frac{0}{0}$. Factor the numerator as a difference of squares: $\lim_{x \to 3} \frac{(x-3)(x+3)}{x-3} = \lim_{x \to 3} (x + 3)$ Step 2 (Evaluate Limit): Substitute $x = 3$: $3 + 3 = \mathbf{6}$ Geometric Interpretation: At $x = 3$, the graph of $f(x)$ has a removable discontinuity (a hole) at point **(3, 6)**; for all other points, the graph is the straight line $y = x + 3$.

19 **[Challenging]** • **Target: Evaluating piecewise functions** • **Cross-Lesson: 4.03 Parametric Equations & Modeling**

Prove the trigonometric identity $\frac{1 - \cos 2\theta}{\sin 2\theta} = \tan\theta$ for all values of θ where both sides are defined.

Step 1: Substitute the double-angle identities $\cos 2\theta = 1 - 2\sin^2\theta$ and $\sin 2\theta = 2\sin\theta\cos\theta$: $\frac{1 - (1 - 2\sin^2\theta)}{2\sin\theta\cos\theta}$ Step 2: Simplify the numerator: $1 - 1 + 2\sin^2\theta = 2\sin^2\theta$ Step 3: Form the ratio and cancel common factor $2\sin\theta$ (for $\sin\theta eq 0$): $\frac{2\sin^2\theta}{2\sin\theta\cos\theta} = \frac{\sin\theta}{\cos\theta} = \tan\boldsymbol{\theta}$ ■

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20 [Challenging] • Target: Checking continuity of piecewise functions • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Solve the exponential equation $4^{x+1} - 5(2^x) + 1 = 0$ for all real values of x.

Rewrite the first term in terms of base 2: $4^{x+1} = 4^1 \cdot (2^2)^x = 4 \cdot (2^x)^2$. Substitute $u = 2^x$ (with constraint $u > 0$): $4u^2 - 5u + 1 = 0$. Factor the quadratic: $(4u - 1)(u - 1) = 0 \Rightarrow u = \frac{1}{4}$ or $u = 1$. Back-substitute $2^x = u$:$2^x = \frac{1}{4} = 2^{-2} \Rightarrow x = -22^x = 1 = 2^0 \Rightarrow x = 0$Both solutions are real and valid: $x = -2$ and $x = 0$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating piecewise functions

For the implicit relation $x^2 + y^2 = 625$, find all y-values when $x = 24$.

a. $y = \pm 25$

b. $y = \pm 24$

c. $y = \pm 7$ ✓

d. $y = 7$

$24^2 + y^2 = 625$: $y^2 = 625 - 576 = 49$, so $y = \pm 7$.

22 [Foundational] • Target: Checking continuity of piecewise functions

A shipping company charges a flat 7 dollars for packages up to 3 kg, plus $\frac{5}{2}$ dollars per kg for anything over 3 kg. Find the cost for a 13 kg package.

a. 32 ✓

b. 33

c. 31

d. $\frac{79}{2}$

$C(13) = 7 + \frac{5}{2}(13 - 3) = 7 + \frac{5}{2}(10) = 32$ dollars.

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23 [Foundational] • Target: Visualizing a piecewise graph

The function f is defined by $f(x) = \frac{2x^2 + kx - 6}{x - 2}$ for all real numbers $xeq2$, where k is a constant. If the graph of $y = f(x)$ in the standard (x, y) coordinate plane has a removable discontinuity (a hole) at the point $(2, L)$, what is the value of L?

a. -1

b. 3

c. 7 ✓

d. 11

A rational function has a removable discontinuity at $x = 2$ if and only if both the numerator and the denominator share the factor $(x - 2)$. Therefore, the numerator must equal 0 when $x = 2$: $2(2)^2 + k(2) - 6 = 0\ 8 + 2k - 6 = 0\ 2k + 2 = 0 \Rightarrow k = -1$ Now, substitute $k = -1$ back into the numerator and factor it: $2x^2 - x - 6 = (x - 2)(2x + 3)$ For all $xeq2$, simplify $f(x)$: $f(x) = \frac{(x - 2)(2x + 3)}{x - 2} = 2x + 3$ The y-coordinate L of the removable discontinuity is the limit of.

24 [Foundational] • Target: Implicit relations and testing for functions

The graph of the sinusoidal function $y = A\sin(Bx + C) + D$ is shown below. The function has a maximum value of 5 and a minimum value of -1. The graph passes through the point $(0, 2)$ and has a period of $\frac{\pi}{2}$. Find the value of A and B. Assume $A > 0$ and $B > 0$.

a. $A = 3, B = 4$ ✓

b. $A = 3, B = 2$

c. $A = 6, B = 4$

d. $A = 6, B = 2$

The amplitude A is half the difference between the maximum and minimum values: $A = \frac{5 - (-1)}{2} = \frac{6}{2} = 3$. Since $A > 0$ is assumed, $A = 3$. The vertical shift D is the average of the maximum and minimum values: $D = \frac{5 + (-1)}{2} = \frac{4}{2} = 2$. The period of the function is given by $T = \frac{2\pi}{|B|}$. We are given that the period is $\frac{\pi}{2}$, so $\frac{2\pi}{B} = \frac{\pi}{2}$ (since $B > 0$). Solving for B, we get $B = \frac{2\pi}{\pi/2} = 4$. Thus, $A = 3$ and $B = 4$.

25 [Foundational] • Target: Implicit differentiation-style reasoning (algebraic slope)

A piecewise function is defined by $f(x) = 4x + 2$ for $x < -3$, and $f(x) = 3x - 4$ for $x \geq -3$. Find $f(-4)$.

a. -15

b. -17

c. **-14** ✓

d. -13

Since $-4 < -3$, use the first piece: $f(-4) = 4(-4) + 2 = -14$.

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26 [Foundational] • Target: Modeling with piecewise functions

A piecewise function is defined by $f(x) = 2x + 3$ for $x < 1$ and $f(x) = x + k$ for $x \geq 1$. Find k so that f is continuous at $x = 1$.

a. 5

b. 6

c. **4 ✓**

d. 1

Continuity: $2(1) + 3 = 1(1) + k \Rightarrow k = 4$.

27 [Foundational] • Target: Piecewise functions with three or more pieces

The graph of a third-degree polynomial function $P(x)$ is shown in the standard (x, y) coordinate plane below. The graph crosses the x-axis at $(-1, 0)$, is tangent to the x-axis at $(2, 0)$, and passes through the point $(0, -2)$. What is the value of $P(-2)$?

a. -16

b. -8

c. 4

d. **8 ✓**

To find $P(-2)$, we first determine the algebraic formula for the polynomial $P(x)$. 1. **Identify the factors and their multiplicities:** - The graph crosses the x-axis at $x = -1$, meaning $(x + 1)$ is a factor with an odd multiplicity. - The graph is tangent to the x-axis at $x = 2$, meaning $(x - 2)$ is a factor with an even multiplicity (at least 2). - Since $P(x)$ is a third-degree (cubic) polynomial, the sum of the multiplicities must equal 3. Therefore, $(x + 1)$ has multiplicity 1 and $(x - 2)$ has mul.

28 [Foundational] • Target: Graphing implicit relations point-by-point

The graph of $g(x) = f(x + 1) - 5$ is obtained from $f(x)$ by:

a. Shifting 1 units left and 5 units down

b. **Shifting 1 units right and 5 units down ✓**

c. Shifting 1 units right and 5 units up

d. Reflecting over the x-axis

Replacing x with $x - 1$ shifts horizontally by 1 (right if positive); adding -5 shifts vertically by -5.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Piecewise functions and average rate of change

Consider a point $P(x, y)$ that lies on the parabola defined by the equation $y = x^2 - 6x + 13$. Let F be the focus of this parabola. If the distance from P to the directrix of the parabola is minimized, what is the y-coordinate of the point P?

a. 3

b. **4** ✓

c. 5

d. 6

The given parabola is $y = x^2 - 6x + 13$. We can rewrite this equation in vertex form by completing the square: $y = (x^2 - 6x + 9) + 13 - 9$, which simplifies to $y = (x - 3)^2 + 4$. This is the equation of a parabola with vertex $(h, k) = (3, 4)$, opening upwards. The standard form of a parabola opening upwards is $(x - h)^2 = 4p(y - k)$, where (h, k) is the vertex and p is the distance from the vertex to the focus and from the vertex to the directrix. Comparing our equation $(x - 3)^2 = 1(y - 4)$ with the s.

30 [Intermediate] • Target: Evaluating piecewise functions

What is the sum of all real values of k for which the equation $|x^2 - 6x + 5| = 2x + k$ has exactly three distinct real solutions?

a. -3 ✓

b. -1

c. -2

d. -13

We consider the piecewise definition of $f(x) = |x^2 - 6x + 5| = |(x - 1)(x - 5)|$: xx = 2 \pm \sqrt{-1 - k} $liestrictlyin$(1, 5), $contributing2solutions$. $-$ Atk = -2, $therootsare$x = 1andx = 3. $Since$x = 1 $isanendpoint$, $only$x = 3$liesin$(1, 5), $contributing1solution$. $-$ Fork \in (-10, -2), $onlythelargerroot2$ + \sqrt{-1 - k}$liesin$(1, 5), $contributing1solution$. 2. * * $Outersections$(x \le 1orx \ge 5) : * * $Setting$ x^2 - 6x + 5 = 2x + kx The roots are $x = 4 \pm \sqrt{11 + k}$. - For.

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31 [Intermediate] • Target: Checking continuity of piecewise functions

The quadratic function $f(x) = ax^2 + bx + c$ has its vertex at $(-1, 5)$. If the graph of $f(x)$ also passes through the point $(2, -13)$, find the value of b.

a. 3

b. -3

c. 6

d. **-6** ✓

The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(-1, 5)$, we have $f(x) = a(x - (-1))^2 + 5 = a(x + 1)^2 + 5$. The graph passes through the point $(2, -13)$, so we can substitute these values into the equation: $-13 = a(2 + 1)^2 + 5$. This simplifies to $-13 = a(3)^2 + 5$, so $-13 = 9a + 5$. Subtracting 5 from both sides gives $-18 = 9a$, which means $a = -2$. Now, we expand the vertex form to find the standard form $f(x) = ax^2 + bx + c$.

32 [Intermediate] • Target: Visualizing a piecewise graph

Consider the function $f(x) = ax^2 + bx + c$. The graph of $y = f(x)$ is a parabola that passes through the points $(-1, 5)$ and $(2, -7)$. If the vertex of the parabola is located at $x = -1/2$, find the value of c.

a. **1** ✓

b. 2

c. 3

d. 4

The vertex of a parabola given by $f(x) = ax^2 + bx + c$ occurs at $x = -b/(2a)$. We are given that the vertex occurs at $x = -1/2$, so $-b/(2a) = -1/2$, which implies $b = a$. The parabola passes through $(-1, 5)$, so $f(-1) = a(-1)^2 + b(-1) + c = a - b + c = 5$. Since $b = a$, this becomes $a - a + c = 5$, which simplifies to $c = 5$. The parabola also passes through $(2, -7)$, so $f(2) = a(2)^2 + b(2) + c = 4a + 2b + c = -7$. Substituting $b = a$ and $c = 5$ into this equation gives $4a + 2a + 5 = -7$, so.

33 [Intermediate] • Target: Implicit relations and testing for functions

Consider the rational function $f(x) = \frac{ax^2 + bx + c}{dx^2 + ex + f}$, where a, b, c, d, e, f are non-zero constants. Given that $\lim_{x \to \infty} f(x) = \frac{1}{2}$ and $\lim_{x \to -1} f(x) = \infty$, determine the value of $\frac{a}{d}$.

a. $\frac{1}{2}$ ✓

b. 1

c. 2

d. Cannot be determined

For a rational function where the degree of the numerator is equal to the degree of the denominator, the limit as $x \to \infty$ is the ratio of the leading coefficients. Therefore, $\lim_{x \to \infty} f(x) = \frac{a}{d}$. We are given that this limit is $\frac{1}{2}$, so $\frac{a}{d} = \frac{1}{2}$. The condition $\lim_{x \to -1} f(x) = \infty$ implies that $x = -1$ is a vertical asymptote. This means that the denominator is zero when $x = -1$, and the numerator is non-zero when $x = -1$. So.

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34 [Intermediate] • Target: Implicit differentiation-style reasoning (algebraic slope)

In the xy-plane, an angle θ is in standard position. If the terminal side of θ passes through the point $(-5, 12)$, what is the value of $\sin(\theta)$?

a. $\frac{12}{13}$ ✓

b. $-\frac{5}{13}$

c. $-\frac{12}{13}$

d. $-\frac{12}{5}$

To find $\sin(\theta)$ for an angle in standard position whose terminal side passes through $(x, y) = (-5, 12)$: 1. Calculate the distance r from the origin to the point: $r = \sqrt{x^2 + y^2} = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ 2. By the coordinate definition of trigonometric functions: $\sin(\theta) = \frac{y}{r} = \frac{12}{13}$ Misconception checks: - $-\frac{5}{13}$ results from confusing sine with cosine, where $\cos(\theta) = \frac{x}{r}$. - $-\frac{12}{13}$ comes from mistakenly as.

35 [Intermediate] • Target: Modeling with piecewise functions

Consider the function $f(x) = ax^2 + bx - 5$. If the graph of $f(x)$ passes through the points $(-2, 3)$ and $(1, -3)$, find the value of $a - b$.

a. 1 ✓

b. -1

c. 2

d. -2

The function is given by $f(x) = ax^2 + bx - 5$. Since the graph passes through $(-2, 3)$, we have $f(-2) = 3$: a(-2)^2 + b(-2) - 5 = 3 \\ 4a - 2b - 5 = 3 \\ 4a - 2b = 8 \\ 2a - b = 4 \quad (1) Since the graph passes through $(1, -3)$, we have $f(1) = -3$: a(1)^2 + b(1) - 5 = -3 \\ a + b - 5 = -3 \\ a + b = 2 \quad (2) Now we have a system of two linear equations with two variables: 1) $2a - b = 4$ 2) $a + b = 2$ Adding equation (1) and equation (2): (2a - b) + (a + b) = 4 + 2 \\ 3a = 6 \\ a = 2 Substituting.

36 [Intermediate] • Target: Piecewise functions with three or more pieces

The function f is defined by $f(x) = -x^2 + bx + c$, where b and c are constants. In the standard (x, y) coordinate plane, the maximum value of $f(x)$ is 14, and $f(0) = 5$. If $b > 0$, what is the value of $f(5)$?

a. 5

b. 10 ✓

c. 13

d. 14

We are given $f(x) = -x^2 + bx + c$. Step 1: Determine the constant c. Evaluating at $x = 0$: $f(0) = -(0)^2 + b(0) + c = 5 \Rightarrow c = 5$ Step 2: Express the maximum value in terms of b. Since the coefficient of x^2 is $-1$0, we have $b = 6$. Step 4: Evaluate $f(5)$. The complete function is $f(x) = -x^2 + 6x + 5$. $f(5) = -(5)^2 + 6(5) + 5 = -25 + 30 + 5 = 10$ A common error is confusing the parameter b with the vertex coordinate x_v (setting $x_v = 6$), which incorrectly leads to.

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37 [Intermediate] • Target: Graphing implicit relations point-by-point

Consider the rational function $f(x) = \frac{x^2 - 9}{x - 3}$. What is the value of $f(x)$ as x approaches 3?

a. 0

b. **6** ✓

c. Undefined

d. 3

The function is given by $f(x) = \frac{x^2 - 9}{x - 3}$. We can factor the numerator as a difference of squares: $x^2 - 9 = (x - 3)(x + 3)$. Thus, for $x \neq 3$, we can simplify the function to $f(x) = \frac{(x - 3)(x + 3)}{x - 3} = x + 3$. To find the limit as x approaches 3, we can substitute 3 into the simplified expression: $\lim_{x \to 3} f(x) = \lim_{x \to 3} (x + 3) = 3 + 3 = 6$. Although the function is undefined at $x = 3$, the limit exists and is 6. Therefore, as x approaches 3, $f(x)$ approaches 6.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Piecewise functions and average rate of change • Cross-Lesson: 4.01 Inverse & Composite Functions

Consider the rational function $f(x) = \frac{x^2 - 4}{x - 2}$. What is the value of the function as x approaches 2?

a. **4** ✓

b. 0

c. Undefined

d. 2

The given function is $f(x) = \frac{x^2 - 4}{x - 2}$. We can factor the numerator as a difference of squares: $x^2 - 4 = (x - 2)(x + 2)$. Therefore, $f(x) = \frac{(x - 2)(x + 2)}{x - 2}$. For $x \neq 2$, we can cancel the $(x - 2)$ term, so $f(x) = x + 2$. As x approaches 2, the value of $f(x)$ approaches $2 + 2 = 4$. Although the function is technically undefined at $x = 2$ due to division by zero, the limit as x approaches 2 exists.

39 [Challenging] • Target: Evaluating piecewise functions • Cross-Lesson: 4.03 Parametric Equations & Modeling

The functions f and g are defined for all real numbers by $f(x) = 2x^2 - 3$ and $g(x) = 3x + 1$ What is the value of $f(g(-2))$?

a. -53

b. 16

c. 47 ✓

d. 95

To find the composite value $f(g(-2))$, evaluate the inner function first, then substitute the result into the outer function: 1. Evaluate $g(-2)$: $g(-2) = 3(-2) + 1 = -6 + 1 = -5$ 2. Substitute -5 into $f(x)$: $f(-5) = 2(-5)^2 - 3 = 2(25) - 3 = 50 - 3 = 47$

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40 [Challenging] • Target: Checking continuity of piecewise functions • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

For $0 < x < 1$, simplify $\cos\left(\sin^{-1}(x)\right)$ as an algebraic expression in x.

a. $\sqrt{1-x^2}$ ✓

b. $\frac{1}{\sqrt{1-x^2}}$

c. $1-x^2$

d. $\sqrt{x^2-1}$

Let $\theta = \sin^{-1}(x)$, so $\sin\theta = x$ and $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, where $\cos\theta \geq 0$. Then $\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-x^2}$.

41 [Challenging] • Target: Visualizing a piecewise graph • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Find the cubic polynomial $P(x)$ with leading coefficient 2 whose graph passes through $(0, 6)$ and has zeros $1, -3$ and $1+i$. But wait — to have real coefficients, if $1+i$ is a root, so is $1-i$. That gives degree 4.

a. $P(x) = 2x^3 - 6$

b. $P(x) = 2(x-1)(x+3)(x-1-i)$

c. $P(x) = 2(x-1)^2(x+3)$

d. **This is impossible for a cubic with real coefficients (need degree ≥4 for those zeros)** ✓

Complex roots of polynomials with real coefficients come in conjugate pairs, so $1+i$ forces $1-i$ as well → at least degree 4. Impossible for cubic.

42 [Challenging] • Target: Implicit relations and testing for functions • Cross-Lesson: 1.03 Rates of Change & Concavity

Consider the rational function $f(x) = \frac{x^2-9}{x-3}$. What is the value of $f(4)$?

a. **7** ✓

b. 1

c. 5

d. 13

To find the value of $f(4)$, substitute $x = 4$ into the function. $f(4) = \frac{4^2-9}{4-3} = \frac{16-9}{1} = \frac{7}{1} = 7$.

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43 [Challenging] • Target: Implicit differentiation-style reasoning (algebraic slope) • Cross-Lesson: 2.01 Exponential Functions & Growth Models

For $f(x) = (x - 4)^3(x + 10)$, describe the behavior of the graph at $x = 4$.

a. crosses the x-axis normally (linear)

b. **crosses the x-axis, flattening (S-shaped) near the zero ✓**

c. touches the x-axis and turns around (does not cross)

d. the graph has a vertical asymptote there

The zero $x = 4$ has multiplicity 3, which is odd, so the graph crosses the x-axis, flattening (S-shaped) near the zero.

44 [Challenging] • Target: Modeling with piecewise functions • Cross-Lesson: 2.02 Logarithmic Functions & Properties

The graph of $y = \sin x$ passes through $(0, 0)$. True or False?

a. Only in Quadrant I

b. False

c. Only for $x > 0$

d. **True ✓**

Applying Trigonometric Functions principles, $\sin(0) = 0$. ✓. Therefore, the verified correct option is True.

45 [Challenging] • Target: Piecewise functions with three or more pieces • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Find the x-coordinate of the hole in the graph of $f(x) = \frac{x^2 + x - 6}{x^2 - 7x + 10}$.

a. x=5

b. **x=2 ✓**

c. No hole exists

d. x=-3

$f(x) = \frac{(x - 2)(x + 3)}{(x - 2)(x - 5)}$; the common factor $(x - 2)$ gives a hole at $x = 2$.

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